



Register No.: 212223230105

Branch & Sem.: AIDS & 5th

Date: 13/09/2025

Sign of the Invigilator:

Subject Code & Subject Name: 19AM401 & Time Series Analysis and Forecasting

6(A) The following data represents the number of sold units yearly at a large department stores:

Year	1992	1993	1994	1995	1996	1997	1998	1999
y_t	1	3	2	4	3	2	3	2

calculate the autocorrelation coefficient (ut to 4 lag) and draw the estimated autocorrelation function.

7) B) Discuss the behavior of ACF and PACF for determining the order of an ARMA Process

8) B) calculate the seasonal indexes using the Simple Average method for the following data.

	Quarter			
Year	I	II	III	IV
2005	71	89	106	78
2006	71	90	108	79
2007	73	91	111	81
2009	76	97	122	89

6) A) Given

Year	1992	1993	1994	1995	1996	1997	1998	1999
y_t	1	3	2	4	3	2	3	2

$$\bar{y} = \frac{20}{8} = 2.5$$

$$s(k) = \frac{\sum_{t=1}^{n-k} (y_t - \bar{y})(y_{t+k} - \bar{y})}{\sum_{t=1}^8 (y_t - \bar{y})^2}$$

$$s(k) = P(k)$$

also, we calculate the pair $(y_t - \bar{y})$

Year	1992	1993	1994	1995	1996	1997	1998	1999
$y_t - \bar{y}$	-1.5	0.5	-0.5	1.5	0.5	-0.5	0.5	-0.5
$(y_t - \bar{y})^2$	2.25	0.25	0.25	2.25	0.25	0.25	0.25	0.25

$$1) \quad s(1) = P(1) = \frac{1}{6} \sum_{t=1}^{8-1} (y_t - \bar{y})(y_{t+1} - \bar{y})$$

$$= \frac{1}{6} \left[(-1.5)(0.5) + (0.5)(-0.5) + (-0.5)(1.5) + (1.5)(0.5) + (0.5)(-0.5) + (-0.5)(0.5) + (0.5)(-0.5) \right]$$

$$= \frac{1}{6} (-0.75 - 0.25 - 0.75 + 0.75 - 0.25 - 0.25 - 0.25)$$

$$= \frac{1}{6} (-1.25)$$

$$s(1) = -0.20$$



$$2) \gamma(2) = P(2) = \frac{1}{6} \sum_{t=1}^{8-2} (y_t - \bar{y})(y_{t+2} - \bar{y})$$

$$= \frac{1}{6} [(-1.5)(-0.5) + (0.5)(1.5) + (-0.5)(0.5) + (1.5)(-0.5) + (0.5)(0.5) + (-0.5)(-0.5)]$$

$$= \frac{1}{6} [0.75 + 0.75 - 0.25 - 0.75 + 0.25 + 0.25]$$

$$= \frac{1}{6} (1)$$

$$\boxed{\gamma(2) = 0.16}$$

$$3) \gamma(3) = P(3) = \frac{1}{6} \sum_{t=1}^{8-3} (y_t - \bar{y})(y_{t+3} - \bar{y})$$

$$= \frac{1}{6} [(-1.5)(1.5) + (0.5)(0.5) + (-0.5)(-0.5) + (1.5)(0.5) + (0.5)(-0.5)]$$

$$= \frac{1}{6} [-2.25 + 0.25 + 0.25 + 0.75 + 0.25]$$

$$= \frac{1}{6} [-1.25]$$

$$\boxed{\gamma_3 = -0.20}$$

$$4) \gamma(4) = P(4) = \frac{1}{6} \sum_{t=1}^{8-4} (y_t - \bar{y})(y_{t+4} - \bar{y})$$

$$= \frac{1}{6} [(-1.5)(0.5) + (0.5)(-0.5) + (-0.5)(0.5) + (1.5)(-0.5)]$$

$$= \frac{1}{6} [-0.75 + 0.25 - 0.25 - 0.75]$$

$$= \frac{1}{6} (-2)$$

$$\boxed{= -0.333}$$

the 4 lags are

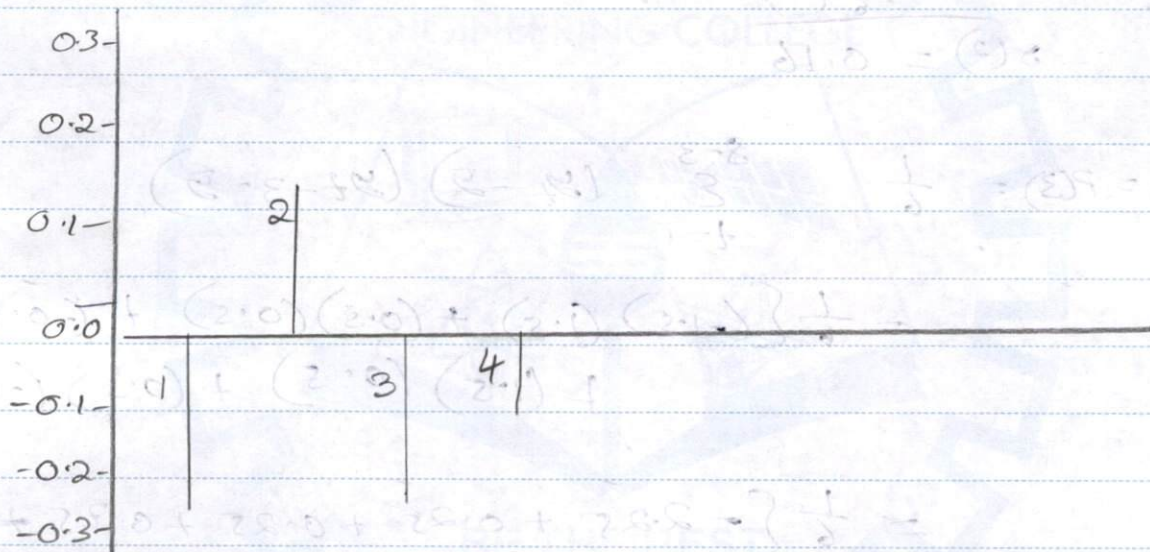
$$\gamma(1) = -0.20$$

$$\gamma(2) = 0.16$$

$$\gamma(3) = -0.20$$

$$\gamma(4) = -0.333$$

These the calculation of the 4 lags now, we will display in the graph with horizontal lines.





7(B)

The behaviour of ACF and PACE for determining the order of an ARMA process

model	ARF ACF	PACF
Autoregressive models $X_t = \mu + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \epsilon_t$	Down time	Cut off lag p
Autoregressive moving model $Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \dots - \theta_p \epsilon_{t-p}$	Cut off lag p	Down time
Autoregressive both X_t and Y_t models $Y_t = \mu + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \epsilon_t - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \dots - \theta_p \epsilon_{t-p}$	Down time	Down time

These are the behaviour of the ACF and PACF. Now we will see the determining order of the ARMA process. In the Time Series and analysis forecasting process the ARMA are so important in calculating determining process. Now we will see the order of the ARMA process.

Model	ΦP	$\Phi K K$
AR(1)	Generate the exponential α in a sumisodial order	cut off object at first time of the lag
AR(2)	Generate the exponential α in a sumisodial order	cut off the object at second time at the lag
AR(P)	Generate the exponential α in a sumisodial order	cut off the time at lag P.
MA(1)	cutt off object at first time at lag	Generate the exponential α in a sumisodial order
MA(2)	cutt off object at second time at lag	Generate the exponential α in a sumisodial order
MA(a)	cutt off at the time lag a	Generate the exponential α in a sumisodial order
ARMA(X-Y)	Generate the exponential at $(X-Y)$ range α in an sumisodial order	Generate the exponential at $(X-Y)$ range α in an sumisodial order



8)

B) Given:

i)

Year	Quarter				Years of Average
	I	II	III	IV	
2005	71	89	106	78	$\frac{71+89+106+78}{4} = 86$
2006	71	90	108	79	$\frac{71+90+108+79}{4} = 87$
2007	73	91	111	81	$\frac{73+91+111+81}{4} = 89$
2009	76	97	122	89	$\frac{76+97+122+89}{4} = 91$

Calculating Years of Average:

~~$$\frac{75+89+106+78}{4}$$~~

We have calculated the Average now we will calculate the Seasonal indexes.

Year	Quarter			
	I	II	III	IV
2005	$\frac{71}{86} \times 100 = 82.5$	$\frac{89}{86} \times 100 = 103.4$	$\frac{106}{86} \times 100 = 123.2$	$\frac{78}{86} \times 100 = 90.6$
2006	$\frac{71}{87} \times 100 = 81.6$	$\frac{90}{87} \times 100 = 103.4$	$\frac{108}{87} \times 100 = 124.1$	$\frac{79}{87} \times 100 = 90.8$
2007	$\frac{73}{89} \times 100 = 82.02$	$\frac{91}{89} \times 100 = 102.2$	$\frac{111}{89} \times 100 = 124.7$	$\frac{81}{89} \times 100 = 91.01$
2009	$\frac{76}{91} \times 100 = 107.01$	$\frac{97}{91} \times 100 = 136.6$	$\frac{122}{91} \times 100 = 177.8$	$\frac{89}{91} \times 100 = 125.3$
Mean	353.16	445.6	543.8	397.71
Total	88.29	111.4	135.95	99.42

The Seasonal Indexes using the Simple Average method for the following data is Seasonal I - 88.29, Seasonal II - 111.4, Seasonal III - 135.95, Seasonal IV - 99.42. These are the data of Seasonal Index.

Seasonal

Seasonal I	Seasonal II	Seasonal III	Seasonal IV
88.29	111.4	135.95	99.42

Calculating the average

$$\frac{88.29 + 111.4 + 135.95 + 99.42}{4}$$

we have calculated the average from the data.

Seasonal

Seasonal I	Seasonal II	Seasonal III	Seasonal IV
88.29	111.4	135.95	99.42